

9 Manifolds[Fenichel's Theorem

(Recall) Singular Perturbation Systems: $0 < \varepsilon \ll 1$
fast $\begin{cases} \dot{u} = f(u, v, \varepsilon) \\ \dot{v} = \varepsilon g(u, v, \varepsilon) \end{cases} \xrightarrow{\tau = \varepsilon t} \text{slow} \begin{cases} \varepsilon u' = f(u, v, \varepsilon) \\ v' = g(u, v, \varepsilon) \end{cases} \xrightarrow{\varepsilon = 0} \text{layer} \begin{cases} \dot{u} = f(u, v, 0) \\ \dot{v} = 0 \end{cases} \xrightarrow{\text{reduced}} \begin{cases} 0 = f(u, v, 0) \\ v' = g(u, v, 0) \end{cases}$
ps: fast var. u (can be vector); slow var. v .
Critical Manifold: $M_0 := \{(u, v) : f(u, v, 0) = 0\}$ ps: reduced problem is defined on M_0
Normal Hyperbolicity: Critical manifold M_0 is normally hyperbolic if:
[$\Re(\lambda_i) \neq 0$ (特征值) of $J = [f_{u1} \ f_{u2} \ \dots]$] $\Rightarrow$ For u 1-dim: If $\frac{\partial f}{\partial u} \neq 0$
If $\Re(\lambda_i) < 0 (> 0) \Rightarrow \frac{\partial f}{\partial u} < 0 (> 0)$: M_0 is **normally attracting** (stable) (**normally repelling** (unstable)).
Fenichel's Theorem: If $M_0 \subseteq \{(u, v) : f(u, v, 0) = 0\}$ is compact (closed & bounded) and normally hyperbolic.
Suppose f, g are smooth. Then for $\varepsilon > 0$ (sufficiently small), $\exists$ Manifold M_ε s.t. it's $\mathcal{O}(\varepsilon)$ close to M_0 and diffeomorphic to M_0 . (i.e. M_ε is locally invariant under the flow of the full system.) M_ε called **slow manifold**.
Find Slow Manifold M_ε : If $M_0 = \{(u, v) : u = h_0(v)\}$, then $M_\varepsilon = \{(u, v) : u = h_\varepsilon(v)\}$
e.g. Given exact sol: $(u, v) = ((u_0 - \frac{v_0^2}{1-2\varepsilon})e^{-t} + \frac{v_0^2}{1-2\varepsilon}e^{-2\varepsilon t}, v_0e^{-\varepsilon t})$. Given $M_0 = \{(u, v) : u = v^2\}$.
Then 1. Linearise, $\tau = \varepsilon t \Rightarrow (u, v) = (\frac{v_0^2}{1-2\varepsilon}e^{-2\tau}, v_0e^{-\tau}) \Rightarrow u = \frac{v^2}{1-2\varepsilon}$. 2. Check, when $\varepsilon = 0, u = v^2$.
3. Therefore, $M_\varepsilon = \{(u, v) : u = \frac{v^2}{1-2\varepsilon}\}$. If $[u]' = [\frac{v^2}{1-2\varepsilon}]'$, it's invariant. (under slow flow)
Invariance of Manifold: $M_\varepsilon = \{(u, v) : u = h_\varepsilon(v)\}$, if $[u]' = [h_\varepsilon(v)]'$, $' = d/d\tau$ 或对应的.

10 Havesting Model

Discrete Population Model: $N_{t+1} = N_t + rN_t(1 - \frac{N_t}{K}) - H_t$, ps: 分析类似 continuous model.
- If $N_{\min} < 1$ **species eradicated/extinct**.
- **Recovery Time**: $T = -\frac{1}{\lambda}$, where $\lambda = f'(N^*)$
Continuous Population Model: $\frac{dN}{dt} = rN(1 - \frac{N}{K}) - h(N)$, r : growth rate, K : carrying capacity.
 $h(N)$: harvesting term. a.constant: $h(N) = Y_0$; b.density-dependent: $h(N) = EN$ (E : harvesting effort).
For the density-dependent harvesting, we have:
- If $E > r$, species eradicated/extinct. If $E < r$, we have the following:
- **Maximum (Sustainable) Yield**: $= \max_N h(N) = h(\frac{K}{2}) = \frac{rK}{4}$ ps: at $N^* = \frac{K}{2}, E = \frac{r}{2}$
Improved our Model: Combine both harvesting types: $h(N) = EN$ for N small, $h(N) = Y_0$ for N large.

A simple model for the concentration $c(x, t)$ of phytoplankton in a cylindrical body of water of length ℓ and cross-sectional area A is given by

$$\frac{\partial c}{\partial t} = D \frac{\partial^2 c}{\partial x^2} + \alpha c. \tag{1}$$

Here, we assume that c depends on the longitudinal position only, i.e. that there is no transverse diffusion; moreover, $D > 0$ is the constant diffusivity, whereas $\alpha > 0$ is the growth rate of plankton. Finally, we impose the boundary and initial conditions

$$c(0, t) = 0 = c(\ell, t) \quad \text{for } t > 0 \quad \text{and} \quad c(x, 0) = c_0(x). \tag{2}$$

(In particular, we hence assume that plankton cannot survive outside of its physiologically suitable domain $[0, \ell]$.)

- (a) Show that the transformation $u(x, t) = c(x, t)e^{-\alpha t}$ reduces Equation (1) to the simple diffusion equation

$$\frac{\partial u}{\partial t} = D \frac{\partial^2 u}{\partial x^2}, \tag{3}$$

subject to

$$u(0, t) = 0 = u(\ell, t) \quad \text{for } t > 0 \quad \text{and} \quad u(x, 0) = c_0(x). \tag{4}$$

[4]

- (b) Solve Equation (3) with boundary and initial conditions as in (4), and hence infer a solution to Equation (1). (*Hint*: make the separation ansatz $u(x, t) = X(x)T(t)$ in (3).) [11]

- (c) Conclude that the minimum domain length ℓ^* beyond which no increase in the concentration of phytoplankton can occur is given by

$$\ell^* = \pi \sqrt{\frac{D}{\alpha}}. \tag{5}$$

[5]

- (d) Given the solution obtained in (b), calculate the total population of phytoplankton in the given body of water. [5]

Solution.

- (a) We calculate

$$\frac{\partial u}{\partial t} = \frac{\partial c}{\partial t}e^{-\alpha t} - \alpha ce^{-\alpha t} \quad \text{and} \quad \frac{\partial^2 c}{\partial x^2} = \frac{\partial^2 u}{\partial x^2}e^{-\alpha t}.$$

Hence,

$$\frac{\partial c}{\partial t} = \frac{\partial u}{\partial t}e^{\alpha t} + \alpha c = D \left(\frac{\partial^2 u}{\partial x^2} e^{\alpha t} \right) + \alpha c;$$

cancelling αc from both sides of the above equation and dividing out the (non-zero) factor of $e^{\alpha t}$, we find (3), as claimed.

- (b) As per the hint, we make the ansatz $u(x, t) = X(x)T(t)$ in (3). Then, substituting in, we find $X\dot{T} = D\dot{X}T$, where the overdot and the prime denote differentiation with respect to t and x , respectively. Rearranging, we have

$$\frac{1}{D} \frac{\dot{T}}{T} = \lambda = \frac{X''}{X},$$

where λ is a (real) constant that depends neither on x nor on t .

We first consider the resulting equation $X'' = \lambda X$ for X , where we note that the given boundary conditions imply $X(0) = 0 = X(\ell)$, giving a *Sturm-Liouville eigenvalue problem*; correspondingly, we have three cases here.

coalesce : 两点合并	eradicated: 根除	membrane potential: 膜电位	unimodal: 单峰	vaccination: 疫苗接种
predator: 捕食者	prey: 被捕食者	trnsriptase: 反转录酶	invades: 入侵	plausible: 似乎合理的
invasion: 入侵	intraspecific competition: 种内竞争		interspecific competition: 种间竞争	
promote: 促进	suppress: 抑制	eliminate: 消除	repelling: 排斥的	
boundary region = trapping region = invariant region = confined region				

Taylor : $\sin(x) \simeq x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots$	$\cos(x) \simeq 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots$	$\tan(x) \simeq x + \frac{x^3}{3} + \frac{2x^5}{15} + \dots$
$\sinh(x) \simeq x + \frac{x^3}{3!} + \frac{x^5}{5!} + \dots$	$\cosh(x) \simeq 1 + \frac{x^2}{2!} + \frac{x^4}{4!} + \dots$	$\tanh(x) \simeq x - \frac{x^3}{3} + \frac{2x^5}{15} - \dots$
Hyperbolic : $\cosh^2(x) - \sinh^2(x) = 1$	$1 + \tanh^2(x) = \sec^2(x)$	$\sinh(x) = \frac{e^x - e^{-x}}{2}$
Integral : $\int \frac{1}{\sqrt{a^2 - x^2}} dx = \arcsin(\frac{x}{a})$	$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \operatorname{arccosh}(\frac{x}{a})$	$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \arctan(\frac{x}{a})$
$\int \frac{1}{\sqrt{a^2 + x^2}} dx = \operatorname{arcsinh}(\frac{x}{a})$	$\int \frac{1}{x^2 - a^2} dx = \frac{1}{a} \operatorname{arctanh}(\frac{x}{a})$	

11 Regular/Singular Perturbation Equation

Singular: If eq like: $\varepsilon y'' + a(x, \varepsilon)y' + b(x, \varepsilon)y = 0$ (i.e. highest der. mult. by ε)
Regular: If eq like: $y'' + a(x, \varepsilon)\varepsilon y' + b(x, \varepsilon)y = 0$ (i.e. highest der. NOT mult. by ε)
Approximation Solution: Assume sol: $y(x; \varepsilon) = y_0(x) + \varepsilon y_1(x) + \varepsilon^2 y_2(x) + \dots$
Substitute into ODE, compare coeff. of $\mathcal{O}(1), \mathcal{O}(\varepsilon), \mathcal{O}(\varepsilon^2)$, etc. 不留 $Y_0 = 0$ 解, 前面已经得到过
- If **singular**, we cannot approximate the other root. $\rightarrow Y = \varepsilon y$, and get new ODE. $\downarrow$
- Assume sol: $Y(x; \varepsilon) = Y_0(x) + \varepsilon Y_1(x) + \varepsilon^2 Y_2(x) + \dots$, substitute, compare $\mathcal{O}(1), \mathcal{O}(\varepsilon), \mathcal{O}(\varepsilon^2)$, etc.
***Uniformly Valid Approximation/Expression**: sol = outer sol + inner sol - matching part
where matching part = section.7 Asymptotic Matching 的结果

12 Appendix

Desingularisation: For $\begin{cases} U' = f(U, V) \\ V' = g(U, V) \end{cases}$ or $X' = \frac{F(X)}{h(X)} \Rightarrow$ Introduce new variable τ s.t.
 $\frac{dt}{d\tau} = U$ or $\frac{dt}{d\tau} = h(X)$ Then $\begin{cases} \frac{dU}{d\tau} = U f(U, V) \\ \frac{dV}{d\tau} = g(U, V) \end{cases}$ or $\frac{dX}{d\tau} = F(X)$
对于证明 $V(U)$ 是 **heteroclinic connection**: 正常证明 + $V(U) \rightarrow V^\pm$ as $U \rightarrow U^\pm$ So that It is a heteroclinic connection from the unstable manifold $W^u(P^-)$ to the stable manifold $W^s(P^+)$. ($P^\pm$ is equilibrium point)

- For $\lambda < 0$, we can write $\lambda = -m^2$, with m real and positive. Then, $X'' + m^2X = 0$ has the auxiliary equation $m^4 + m^2 = 0$, which gives $m = \pm im$ and, hence,

$$X(x) = c_1 \cos(mx) + c_2 \sin(mx)$$

for some constants c_1 and c_2 that remain to be determined. Imposing the boundary conditions, we have $X(0) = 0 = c_1$ and, hence, $c_1 = 0$. Then, $X(\ell) = 0 = c_2 \sin(m\ell)$ implies $m\ell = n\pi$ or $m = \frac{n\pi}{\ell}$, for $n = 1, 2, \dots$. Therefore, we find that $X(x) \propto \sin(\frac{n\pi}{\ell}x)$ gives a solution for any integer n .

- For $\lambda = 0$, we have $X'' = 0$, which implies $X' = c_1$ or $X(x) = c_1x + c_2$, where c_1 and c_2 are constants that remain to be determined. Imposing again the boundary conditions, we have $X(0) = 0 = c_2$ and $X(\ell) = 0 = c_1\ell + c_2$. Then, it follows that $c_1 = 0 = c_2$, which only gives the trivial (zero) solution. Hence, that case is irrelevant.
- For $\lambda > 0$, we write $\lambda = m^2$, with m real and positive, which yields $X'' - m^2X = 0$, with auxiliary equation $\mu^2 - m^2 = 0$. Hence,

$$X(x) = c_1 \cosh(mx) + c_2 \sinh(mx),$$

where we now need to determine c_1 and c_2 : imposing again the boundary conditions, we obtain $X(0) = 0 = c_1$ and, hence, $X(\ell) = c_2 \sinh(m\ell)$, which implies $c_1 = 0 = c_2$. Hence, that case also gives no non-trivial solution.

It remains to solve $\ddot{T} = \lambda DT$ for T in the only relevant case of $\lambda = -m^2 = -\frac{n^2\pi^2}{\ell^2}$: clearly,

$$T(t) \propto e^{-\frac{n^2\pi^2}{\ell^2}Dt},$$

which in sum implies that

$$X(x)T(t) = \gamma_n \sin\left(\frac{n\pi}{\ell}x\right)e^{-\frac{n^2\pi^2}{\ell^2}Dt}$$

is a solution to (3) for any constant γ_n and $n = 1, 2, \dots$. By linearity, the superposition principle shows that

$$u(x, t) = \sum_{n=1}^{\infty} \gamma_n e^{-\frac{n^2\pi^2}{\ell^2}Dt} \sin\left(\frac{n\pi}{\ell}x\right) \tag{6}$$

gives the general solution.

It remains to determine γ_n for $n = 1, 2, \dots$; to that end, we make use of the given initial condition, which yields

$$u(x, 0) = c_0(x) = \sum_{n=1}^{\infty} \gamma_n \sin\left(\frac{n\pi}{\ell}x\right)$$

and, hence,

$$\gamma_n = \frac{2}{\ell} \int_0^\ell c_0(x) \sin\left(\frac{n\pi}{\ell}x\right) dx. \tag{7}$$

(In other words, γ_n are the Fourier sine coefficients of the function $c_0(x)$ on $[0, \ell]$.)

Finally, rewriting (6) in terms of $c(x, t)$, we have

$$c(x, t) = u(x, t)e^{\alpha t} = \sum_{n=1}^{\infty} \gamma_n e^{(\alpha - \frac{n^2\pi^2}{\ell^2}D)t} \sin\left(\frac{n\pi}{\ell}x\right). \tag{8}$$

- (c) For the concentration of phytoplankton to increase in time, at least one of the exponents $\alpha - \frac{n^2\pi^2}{\ell^2}D$ has to be positive; in other words, $\alpha > \frac{n^2\pi^2}{\ell^2}D$ must hold for some $n = 1, 2, \dots$. Rearranging, we find that the minimum domain length ℓ for which $\ell^2 > n^2\pi^2 \frac{D}{\alpha}$ is found for $n = 1$, which implies

$$\ell > \ell^* = \pi \sqrt{\frac{D}{\alpha}},$$

- (d) The total population $N(t)$ of plankton is obtained by integrating $c(x, t)$, as given in (8), over the volume of the given body of water:

$$\begin{aligned} N(t) &= A \int_0^\ell c(x, t) dx \\ &= A \int_0^\ell \sum_{n=1}^{\infty} \gamma_n e^{(\alpha - \frac{n^2\pi^2}{\ell^2}D)t} \sin\left(\frac{n\pi}{\ell}x\right) dx \\ &= A \sum_{n=1}^{\infty} \gamma_n e^{(\alpha - \frac{n^2\pi^2}{\ell^2}D)t} \int_0^\ell \sin\left(\frac{n\pi}{\ell}x\right) dx. \end{aligned}$$

(Here, we recall that A denotes the cross-sectional area of the given cylinder.) Since

$$\int_0^\ell \sin\left(\frac{n\pi}{\ell}x\right) dx = -\frac{\ell}{n\pi} \cos\left(\frac{n\pi}{\ell}x\right) \Big|_0^\ell = \frac{\ell}{n\pi} [1 - (-1)^n] = \frac{\ell}{n\pi} \begin{cases} 2 & \text{for } n = 2k + 1 \text{ odd,} \\ 0 & \text{for } n = 2k \text{ even,} \end{cases}$$

where $k = 0, 1, 2, \dots$, we can write

$$N(t) = \frac{2A\ell}{\pi} \sum_{k=0}^{\infty} \frac{\gamma_{2k+1}}{2k+1} e^{(\alpha - \frac{(2k+1)^2\pi^2}{\ell^2}D)t}$$

for the total population.